



Chapter Two

Basic Concepts of Logic

Unexamined life isn't worth living.
Socrates.

What is Logic?

- Are you/we logical?
- What does mean to be logical?
- What is the link between logic and being human?
- What do we believe and why do we believe it?

Climbing the ladder



Logic defined...

- **Logic** can be defined as the science that evaluates arguments based on rules and principles of correct reasoning.
- **Logic** is the study of methods for evaluating whether the premises of an argument adequately support its conclusion.
- What is an argument?
 - An argument, as it occurs in logic, is a group of statements, one or more of which (the premises) are claimed to provide support for, or reasons to believe, one of the others (the conclusion).
 - Arguments could be **good or bad**.

Cont'd

- An **argument** is a set of statements where some of the statements, called the *premises*, are intended to support another, called the conclusion.
- A **statement** is a sentence that is either true or false – in other words – typically a declarative sentence or a sentence component that could stand as a declarative sentence.
- Unlike statements, many sentences can't be said either true or false.

Cont'd

- What is the weight of carbon? (question)
- Lets go to the park today. (Proposal)
- We suggest that you travel by bus (suggestion)
- Turn to the left at the next corner (Command)
- Right on! (Exclamation)

Cont'd

- Statements
 - Political candidates always tell the complete truth.
 - No wives ever cheat on their husbands.
 - Orange is a fruit.
 - Barack Obama was the 44th president of the United States of America.
 - The battle of Adwa was fought in 1896.

Conditional Statements

- A **conditional statement** is an “if . . . then . . .” statement.
- Every conditional statement is made up of two component statements.

Conditional statements

Antecedent Consequent

If _____ then _____ .

Consequent Antecedent

_____ if _____ .

Cont'd

- E.g If Sarah Palin loves shooting wolves from airplanes, then she has little respect for wildlife.
- If you study hard then your will score an “A” in logic.
- If you jump off this building then your leg will be broken.
- Finally, while no single conditional statement is an argument, a conditional statement may serve as either the premise or the conclusion (or both) of an argument, as the following examples illustrate:

Cont'd

- If Iran is developing nuclear weapons, then Iran is a threat to world peace.
Iran is developing nuclear weapons.
Therefore, Iran is a threat to world peace.

- Explanation:
- Reports
- Warning
- A piece of advice
 - You should keep a few things in mind before buying a used car. Test drive the car at varying speeds and conditions, examine the oil in the crankcase, ask to see service records, and, if possible, have the engine and power train checked by a mechanic.
- A statement of belief or opinion
 - I think a nation such as ours, with high moral traditions and commitments, has a further responsibility to know how we became drawn into this conflict, and to learn the lessons it has to teach us.
- Expository passages

Illustrations

Arguments vs. Non-arguments

- Not every set of statements is an argument. Not all passages contain arguments.
- An argument consists of:
 - Premises
 - Conclusion
 - An inferential link
- Passages which fail to consist these elements are non-argumentative.

Cont'd

- **Premise:** is part of an argument that provides support or evidence for what is claimed by the arguer.
- **Conclusion:** is what is drawn or follows or is implied by the given evidence.
- **Inferential link:** is the connection between the premise and the conclusion.

How to differentiate a premise from a conclusion

- By examining the role of the statement within the given argument.
- By look at indicator words.

Some typical **conclusion indicators** are

therefore	accordingly	entails that
wherefore	we may conclude	hence
thus	it must be that	it follows that
consequently	for this reason	implies that
we may infer	so	as a result

Cont'd

indicator. Some typical **premise indicators** are

since

in that

seeing that

as indicated by

may be inferred from

for the reason that

because

as

in as much as

for

given that

owing to

Arguments

- **All film stars are celebrities.
Halle Berry is a film star.
Therefore, Halle Berry is a celebrity.**
- All four-legged creatures have wings.
All spiders have four legs.
Hence, all spiders have wings.
- All human beings are immortal.
Jonathan is a human being.
Thus, Jonathan is immortal.

Cont'd

- Every law is evil, for it is an infraction of liberty.
- Expectant mothers should never use recreational drugs as the use of such drugs would jeopardize the growth of the fetus.
- Some birds are mammals. Ostrich is a bird. Thus, ostrich is a mammal.
- All birds have beaks. Some cats are birds. So, some cats have beaks.

What makes an argument good or bad?

- **IN LOGIC, IT IS ALL ABOUT EVIDENCE.**
- **In logic you cannot avoid or skip the question “why?”**
- Every claim needs to be supported by an evidence.
- For an argument to be a good argument, there has to be a good evidence in support of the claim.
- If the premises of an argument do not support the conclusion, even though they are claimed to, the argument is not a good one.

Cont'd

- Arguments can be classified into two main types:
 - **Deductive and Inductive**
 - How is it possible to distinguish a deductive argument from an inductive argument?

Criteria to distinguish a deductive from an inductive argument

- **Definition/ Truth value**
- **Indicator words**
- **Actual support of the premises to the conclusion**
- **Form of argumentation**

Definition/Truth Value

- A **deductive argument** is an argument incorporating the claim that it is *impossible* for the conclusion to be false given that the premises are true. **Deductive arguments are those that involve necessary reasoning.**
- On the other hand, an **inductive argument** is an argument incorporating the claim that it is *improbable* that the conclusion be false given that the premises are true. **Inductive arguments involve probabilistic reasoning.**

Indicator words

- **Deductive indicators:** certainly, definitely, necessarily, absolutely
- **Inductive indicators:** probably, improbably, plausibly, likely, unlikely.
- The *actual* strength of the inferential link between premises and conclusion
- **Necessity** vs. Probability

Forms of Argumentation

- **Deductive Forms**

- A. Argument based on Mathematics**

Such arguments include simple arithmetic and geometrical arguments.

They don't include statistical arguments as they entertain probability.

For example: Nahom has placed three apples and five oranges in this paper bag. Thus, this paper bag consists eight piece of fruits.

Cont'd

- The body temperature of a healthy person is 37°C. Eyasu's father body temperature is currently 42°C. Thus, Eyasu's father is not feeling well.

B. **Argument based on definition**

This is a form argument in which a conclusion is drawn from a definition of a word in the premises.

E.g God is omnipresent. It follows that, God can be anywhere anytime.

Victoria is a prostitute. Thus, Victoria accepts money in exchange for sex.

Cont'd

- C. **Categorical Syllogism:** a categorical syllogism is a deductive argument with two premises and one conclusion. Either the premises or the conclusion consists of a quantifier or a category indicator.
- E.g All medical doctors are graduates of medical school.

Some medical doctors are unethical persons.

Thus, Some unethical persons are graduates of medical schools.

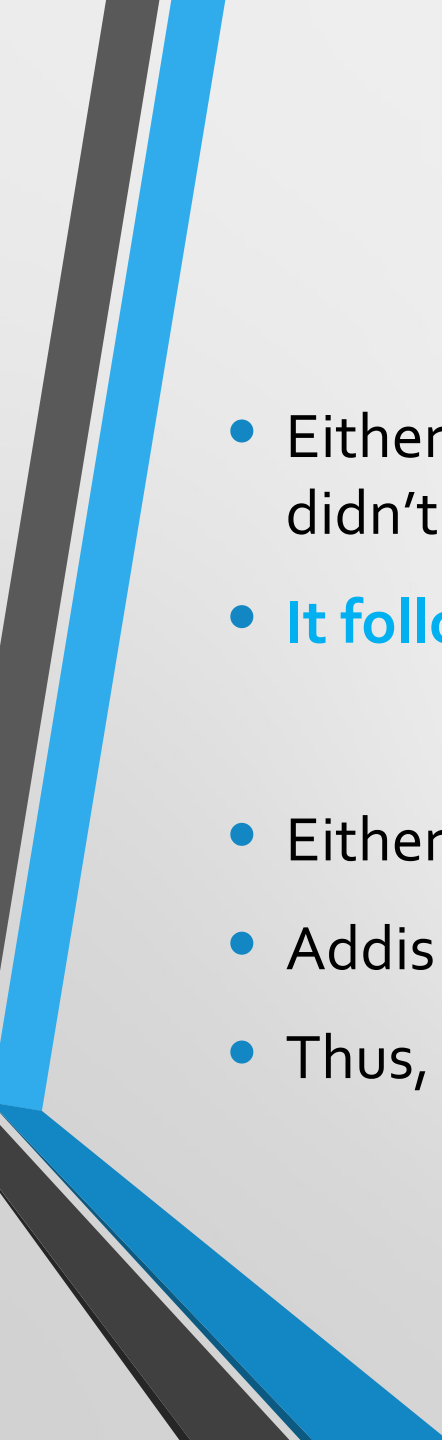
E.g All ancient forests are sources of wonder. Some ancient forests are targets of the timber industry. Therefore, some sources of wonder are targets of the timber industry.

- **D. Hypothetical syllogism:** is a deductive argument consisting of conditional statements either as a premise or conclusion.
- If quartz scratches glass then quartz is harder than glass.
Quartz scratches glass
Thus, quartz is harder than glass.

If Ephrem is a president then he lives in the national palace.
Ephrem is not the president.
Thus, Ephrem does not live in the national palace.

Cont'd

- E. **Disjunctive Syllogism**: is an argument that joins two options by using “either...or” and affirms one of the alternatives by eliminating the other.
- E. g Either breach of a contract is a crime or it isn't punishable by the law.
- Breach of a contract isn't a crime.
- Thus, it is not punishable by the law.

- 
- Either Ethiopia or Italy have won the battle of Adwa. Italy didn't win the battle of Adwa.
 - **It follows that, Ethiopia have won the battle of Adwa.**
 - Either Addis Ababa or Accra is the capital city of Ethiopia.
 - Addis Ababa isn't the capital city of Ethiopia.
 - Thus, Accra is the capital city of Ethiopia.

Inductive forms

- **A. Argument based on prediction:** is an argument that draws a conclusion about the future based on our knowledge of the past or the present.
- E.g The Ethiopian and Nigerian national football teams have played twelve matches and Nigeria have won 8 of these matches while Ethiopia won only 3. The last game they played ended with a draw. Hence, Nigeria will win the 13th match.
- All previous elections in Ethiopia were followed by post election violence. Thus, the next Ethiopian election will also be followed by post election violence.

- **B. Argument based on analogy:** in such arguments a conclusion is drawn from the similar aspects of things, places, or people.
- **E. g** Jonathan's motorbike is blue in color, speeds up to 180km/h and is made in Germany. Ermias's motorbike is blue in color and speeds up to 180km/h. Thus, Ermias's motorbike is also made in Germany.
- My sister usually wears a black dress and hold a blue umbrella. The woman coming from the other side of the street is wearing a black dress and holding a blue umbrella. Thus, the woman coming from the other side of the street is my sister.

- **C. Inductive generalization:** in such arguments a conclusion is drawn from our knowledge about the sample.
- **E.g** This barrel contains 100 apples and eight randomly selected were found to be ripe. Thus, all the 100 apples are ripe.
- **Five of the law books I read are written by philosophers from Greece. Hence, every law book is written by Greek philosophers.**

Cont'd

- **D. Argument based on authority:** is an argument where the conclusion is developed based on the statement of the authority.
- E.g Mike has committed a murder because an eye witness has testified against him in the court of law.
- **E. Argument based on Sign:** this types arguments appeal to the meaning of different signs to draw a conclusion.
- E.g **The package material says “keep it far from children.” So, the package must consist of a medicine.**

Cont'd

- **F. Causal inference/ Causation** : is an argument that draws a conclusion based on the link between a cause and an effect.
- **E.g** The clouds are getting darker and the thunder is roaming. Let's go home quickly; the rainfall is inevitable.

Cont'd

- **Particular to General**
- Three is a prime number.
Five is a prime number.
Seven is a prime number.
Therefore, all odd numbers between two and eight are prime numbers.
- **General to Particular**
- All emeralds previously found have been green.
Therefore, the next emerald to be found will be green.

Valid and Invalid

- Deductive arguments can be evaluated as valid or invalid.
- There are three main criteria through which we can distinguish a valid argument from an invalid one.
 - i. **Definition/truth value**
 - ii. **Structure**
 - iii. **Refutability**

Definition

- A **valid deductive argument** is an argument in which it is impossible for the conclusion to be false given that the premises are true. In these arguments the conclusion follows with strict necessity from the premises.
- Conversely, an **invalid deductive argument** is a deductive argument in which it *is* possible for the conclusion to be false given that the premises are true. In these arguments the conclusion does not follow with strict necessity from the premises, even though it is claimed to.

Structure

All A are B

All B are C

Thus, All A are C

(Valid form)

All A are B

All C are B

Hence, All A are C

(Invalid form)

Refutability

- The important question here is: is it possible to reject the conclusion given the premises or the evidence?
- Is it possible to replace the conclusion given the evidence?

Cont'd

- E.g All wines are beverages.

Chardonnay is a wine.

Therefore, chardonnay is a beverage.

All wines are soft drinks

Mirinda is a wine.

Therefore, Mirinda is a soft drink.

Cont'd

- If Bill Gates owns all the gold in Fort Knox then Bill gates would be wealthy. Bill Gates does not own all the gold in Fort Knox.

Hence, Bill Gates wouldn't be wealthy.

If Ephrem is a president is then he lives in the national palace. Ephrem isn't a president.

Thus, Ephrem doesn't live in the national palace.

Cont'd

- All banks are financial organizations.

Wells Fargo is a bank.

Therefore, Wells Fargo is a financial organization.

Valid

All banks are financial organizations.

Wells Fargo is a financial organization.

Thus, Wells Fargo is a bank.

Invalid

Cont'd

- All financial organizations are banks.

Wells Fargo is a bank .

Thus, Wells Fargo is a financial organization.

Invalid

Cont'd

- All bachelors are persons.

All unmarried men are persons.

Hence, all bachelors are unmarried men.

All A are B

All C are B

Thus, All A are C

Valid

- Its conclusion is true by definition. And also refutability.

Cont'd

All auto makers are computer manufacturers.

Union Carbide is an auto maker.

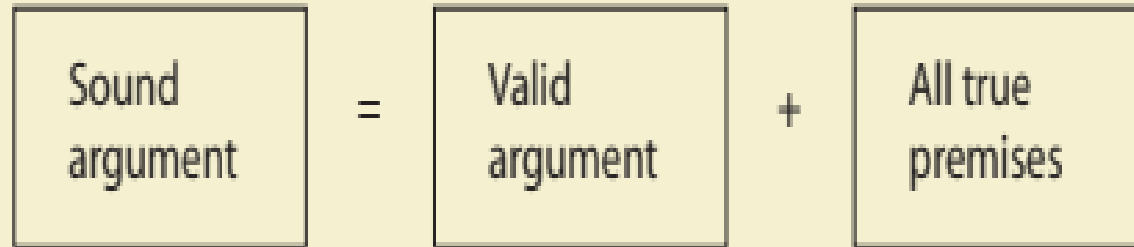
Thus, Union Carbide is a computer manufacturer.

All television networks are media companies. NBC is a television network

Therefore, NBC is a media company.

Sound and Unsound

- A deductive argument can also be evaluated as sound and unsound.



- If one of the two conditions are not met then the argument can be labeled as unsound deductive argument.

Strong and Weak

- A **strong inductive argument** is an inductive argument in which it is **improbable** that the conclusion be false given that the premises are true.
- On the other hand, a **weak inductive argument** is an argument in which the conclusion does not follow probably from the premises, even though it is claimed to.
- There two main criteria to differentiate inductive strong from an inductive weak arguments.

Cont'd

- I. **Truth value**
- II. **Actual support of the premise to the conclusion.**
- In general the strength or weakness of an inductive argument results not from the actual truth or falsity of the premises and conclusion, but from the probabilistic support the premises give to the conclusion.
- N.B The important question is, “how much of a chance does the conclusion have given the premises?”

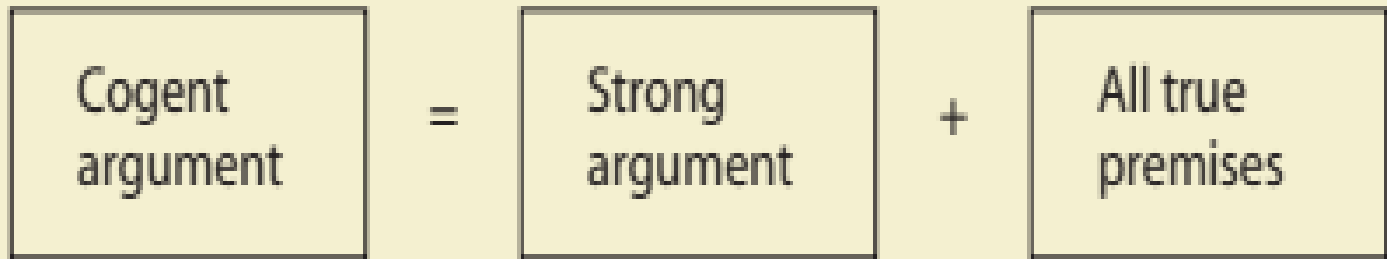
Examples

- All previous American presidents were men. Thus, the next American president will be a man.
- Few American presidents were republicans. Hence, the next American president will be a republican.
- This barrel contains one hundred apples and eighty randomly selected were found to be ripe. Therefore, all the one hundred apples are ripe.

Cont'd

- This barrel contains one hundred apples and three randomly selected were found to be ripe. Therefore, all the one hundred apples are ripe.
- Tigers are striped. Zebras are striped. Tigers are carnivores. Hence, zebras are carnivores as well.
- All dinosaur bones discovered to this day have been at least 50 million years old. Therefore, probably the next dinosaur bone to be found will be at least 50 million years old.

Cogent and Uncogent



- Swimming in the Caribbean is usually lots of fun. Today the water is warm, the surf is gentle, and on this beach there are no dangerous currents. Therefore, it would be fun to go swimming here now



Chapter Three

Logic and Language

Introduction

- ① **Language and its application is one of the fundamental areas of the study of logic. For that matter this chapter focuses on meanings and definitions.**
- ② Ordinary Language serves various functions in our day-to-day lives. The contemporary philosopher Ludwig Wittgenstein thought the number of these functions to be virtually unlimited. They are important to ask questions, to tell lies, to form a hypothesis, to tell jocks, to sing songs, to issue commands...
- ③ To fit with our purpose, we select three of the main linguistic functions:
 - 1. To convey information**
 - 2. To express or evoke feelings.**
 - 3. Directive functions**

Functions of Language

- Emotive function of language is all about expressing feelings. For Example:

She is smart!

I hate him!

What an ugly cat!!

- Directive function of language is used to pass direction or give instructions. For example:

Leave me alone.

Stop drinking.

Give me the key.

Cont'd

- The cognitive aspect of language includes such functions as delivering information and communicating meanings. For example:

Addis Ababa is the capital city of Ethiopia.

Barack Obama is the 44th president of the United States of America.

Orange is a fruit.

Language and Meaning

- The basic units of any ordinary language are words. Though words in general are primary units of ordinary languages, we in this chapter, are not mainly concerned with them, rather with terms. A **term** is a word or group of words, which can be used as the *subject of a statement*. **Proper names, common names and descriptive phrases** can be considered as *terms*. The followings are some of the examples, which are used as terms.

Cont'd

- On the other hand, **verbs, adjectives, prepositions, conjunctions** and the like cannot be considered as terms since they cannot serve as subjects of statement. For example, words such as; 'however', 'argue fairly', 'controversial'...cannot be taken as terms.

Cont'd

- ◎ Terms are made up of words having two kinds of meanings: **intensional** and **extensional**. The intensional meaning contains *the attributes or qualities, which the term connotes* where as extensional meaning consists of *members of the class that the term denotes*. And, **extensional meaning** is usually termed as **extension** or **denotation**.

Intensional Meaning

Philosopher is the one who is capable of being critical reflective and rational.

Water is an odorless compound made up of hydrogen and oxygen and quench ones thirst when drank.

Extensional meaning

- Philosopher is person like Plato, Socrates, Aristotle, Hobbes and Nietzsche.
- Mountain is places like, Alps, Everest, Kilimanjaro, Ras Dashen, Batu, and so on.

The Law of Inverse Variation

- This implies that the **intensional meaning of a term is conversely or indirectly related with its extensional meaning.**
- Decreasing intension /Increasing extension
Biology, Natural sciences, Sciences

Cont'd

- Increasing extension/decreasing intension.
Tiger, cat, mammal, animal

Increasing Intension/decreasing extension.
Human beings, artists, reggae artists

Empty Extension

⦿ Causes of empty extension:

- When the class disappears or vanishes.
- When the term represents no class or is merely the result of human imagination.

For example: “unicorn,” “leprechaun,” “gnome,” “elf,” and “griffin.”

Satan’, ‘Unicorn’, ‘God’ These terms have no classes, members or categories which are included in them.

Definitions and Their Purposes

- Over the years philosophers have held various conflicting views about the purpose of definitions.
- A definition: ***Definiendum*** and ***definiens***.
- ***Definiendum*** is the word, which is supposed to be defined, and **definiens** is a word or group of words, which does the defining. Definition = Definiendum (Word to be defined) + Definiens (Words that do the defining).

Cont'd

- For example, in the Definition “‘Tiger’ means a large, striped, ferocious feline indigenous to the jungles of India and Asia,” the word “tiger” is the definiendum, and everything after the word “means” is the definiens.

Lexical Definition

- Lexical definition is usually called dictionary definition since it is used to report the meaning that the subject to be defined has in a language. Unlike stipulative definition, lexical definition can be evaluated as true or false with reference to the actual usage of the word. And, a good lexical definition should avoid or resist ambiguous meanings.

Cont'd

- Here it should be underlined that **ambiguity** and **vagueness** are two different notions. A word is said to be vague if it **lacks precision** of application. words such as love, happiness, rich, normal... are vague since they are employed in different situations so that they might have different meanings. On the other hand, a word is said to be ambiguous when it contains **two distinct meanings** in that particular situations. Words like 'Sound', 'light', 'right' bank...are ambiguous words.

Stipulative Definition

- Stipulative definition **assigns meanings to words for the first time** either through coining new words or assigning new meanings to old words. The prime function of stipulative definition is to substitute a more complex expression by simpler one. And, it is often caused by some new phenomenon or development. Example, “Tigon” was taken to mean the offspring of a male tiger and a female lion, and “liger” the offspring of a male lion and a female tiger.

Precising Definition

- A précising definition is primarily intended to reduce the **vagueness** of the word. This helps us to reach a decision to the applicability of the word to a specific situation. As we saw in the first section of this chapter, an expression is vague if there are **borderline cases** in which it is impossible to tell if the word applies or does not apply. Words such as “fresh,” “rich,” and “poor” are vague.

- The definition “‘Poor’ means having an annual income of less than \$4,000 and a net worth of less than \$20,000” is an example of a precisising definition.
- *“Antique” means, at least 100 years old.*

Theoretical Definition

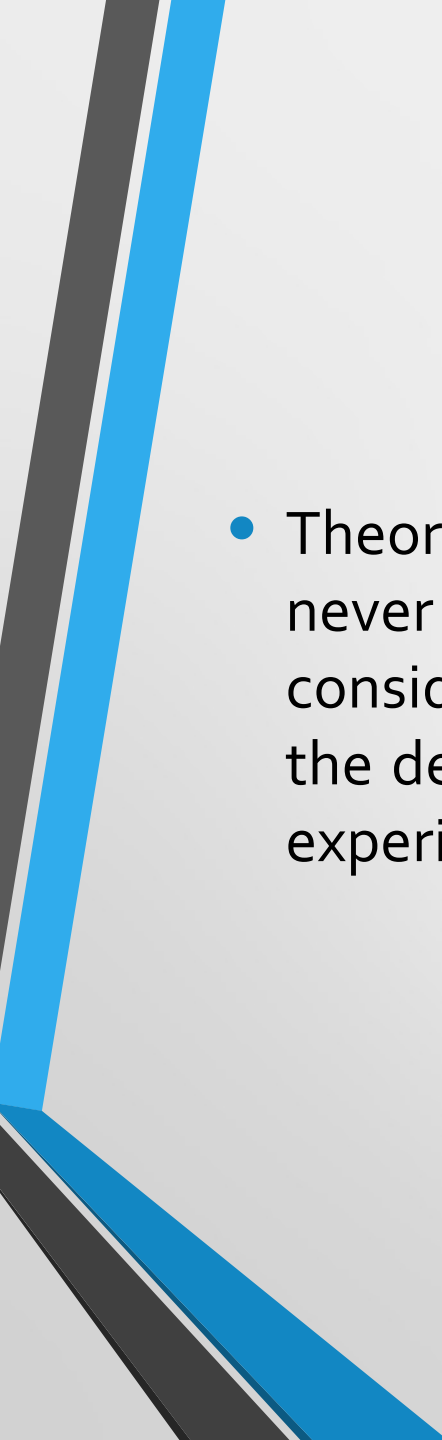
- A theoretical definition assigns a meaning to a word by suggesting a theory that gives a certain characterization to the entities that the term denotes. Such a definition provides a way of viewing or conceiving these entities that suggests deductive consequences, further investigation (experimental or otherwise), and whatever else would be entailed by the acceptance of a theory governing these entities.

Cont'd

- The definition of the term **"heat"** found in texts dealing with the kinetic theory of heat provides a good example: "'heat' means the energy associated with the random motion of the molecules of a substance." This definition does more than merely assign a meaning to a word; it provides a way of conceiving the physical phenomenon that is heat.

Cont'd

- Not all theoretical definitions are associated with science. Many terms in philosophy, such as “substance,” “form,” “cause,” “change,” “idea,” “good,” “mind,” and “God,” have been given theoretical definitions. For example: John Stuart Mill’s definition of “good” as the greatest happiness of the greatest number provided the underpinnings for his utilitarian theory of ethics.

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- Theoretical definitions, like stipulative definitions, can never be evaluated as true or false; however, they can be considered as more interesting or fruitful depending on the deductive consequences they imply and the result of experiment they suggest.

Persuasive Definition

- ① A persuasive definition is characterized through assigning emotively charged meaning to the definiendum as if it were having such meaning in the language in which it is used. This would create positive or negative attitude toward the subject to be defined.
- ② Finally persuasive definitions can be evaluated as true or false though that is not the primary concern here.

For example:

- “Abortion” means the ruthless murdering of innocent human beings.
- “Abortion” means a safe and established surgical procedure whereby a woman is relieved of an unwanted burden.

Techniques of Definition

- Intensional/Connotative
- Extensional/Denotative

Intensional techniques of definition

An **Intensional definition** is one that assigns a meaning to a word by indicating the qualities or attributes that the word connotes.

- **Synonymous Definition**

It is a kind of intentional definition that the definiens is a synonym of the word being defined.

For Example:

- *"Obese" Means fat*
- *"Skinny" means thin.*
- *"Physician" means doctor.*
- *"Intentional" means willful.*

● Etymological Definition

It assigns meanings to a word through showing its root, origin or ancestors in both its own language as well as other languages. Most of the time, English words have ancestors in other language such as Greek, Latin, French and others.

For example: ***"Philosophy" is a word derived from the Greek philia, which means love, and Sophia, which means wisdom.***

Ethics derives from the Greek word 'ethos', which means "custom" in singular and character in plural (ethe).

Operational Definition

- It gives meaning to a word through setting experimental procedures, which will be applied to it. It describes an operation to be performed.
- Examples: - *A knife is "Sharp" if it produces a thin scratch when very gently drawn over one's thumbnail.*
- - *One substance is "harder than" another if and only if one scratches the other when the two are rubbed together.*

Definition by Genus and Differentia

- It assigns a meaning to a term by identifying a genus term and indicating the unique aspect a term from other terms. It is more applicable and adequate definition than any other intensional definitions.
- Ice" means frozen water.
- Daughter" means female offspring.
- "Husband" means married man.

Extensional Techniques

a. **Demonstrative /Ostensive**

This is definition by showing.

b. **Enumerative**

“State” means something such as Ohio, Arkansas, Minnesota, and Tennessee.

“Actress” means a person such as Nicole Kidman, Emma Thompson, or Natalie Portman.

“Continent” means something such as Arica, Asia, North America, Europe and South America.

c. **Definition by sub-class**

“Fictional work” means either a poem, a play, a novel, or a short story.

“Plant” means something such as a tree, a flower, a vine, or a cactus.

Rules for a (good) Lexical Definitions

- As it has been underlined before, lexical definition reports the meaning of a word in a given language and most dictionary definition are lexical definitions . Thus, the followings are some of the rules and requirements, which a lexical definition should satisfy:

- **Proper Grammar**

Spelling, Italicization or quotation marks.

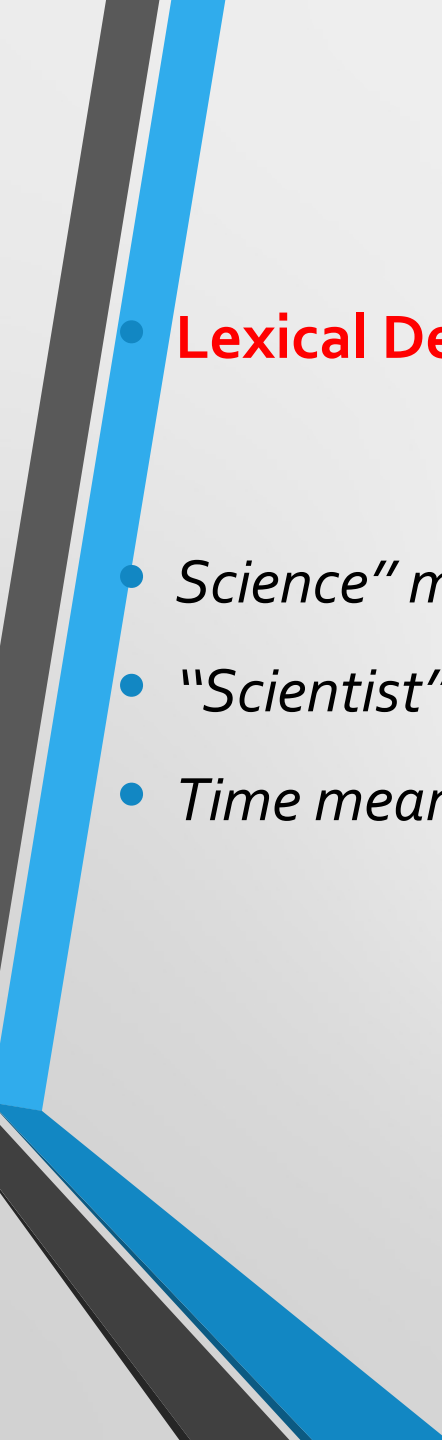
The definiedum should be italicized or put in quotation marks.

Example: *"Judge" means who decides if you go to jail or pay someone pays you or your pay a fine or get of free.*

- **Lexical Definition Should Convey the Essential Meaning of the Word Being Defined.**
- *Human beings* are laughing animals.
- *Human beings* are rational animals.

- *Solider* is a person ready to kill or to be killed.
- *Solider* is a person who is a member of an army.

- **Lexical Definition Should Be Neither Too Broad nor Too Narrow.**
- **Horse is a mammal. (Broad)**
- **Horse is an adult mammal used by cavalry forces. (Narrow)**
- *"Musician" means a person who plays the violin. (Narrow)*

- 
- **Lexical Definition Should Avoid Circularity**
 - *Science* means the activity engaged in by scientists.
 - *"Scientist"* means anyone who engages in science.
 - *Time* means that which is measured by a clock.

- **A Lexical Definition Should Not Be Negative When It Can Be Affirmative**

For example:

- *“Concord” means harmony (positive defn)*
- *-‘Concord” means the absence of discord (negative defn).*
- *However, some terms cannot be defined affirmatively or positively. For example: “Darkness” means the absence of light.*

- **Lexical Definition Should Avoid Figurative, Obscure, Vague, or Ambiguous Language**

Examples of figurative definitions:

"Politics" is the land of Liberty.

Camel is a ship of the desert.

- *Obscure definitions:*

"Tooth brush" means a setaceous dental particle desolator.

- **Lexical Definition Should Avoid Affective Terminology.**

A lexical definition should avoid affective expressions and terminologies.

- "Theism" *means belief in that great Santa Claus in the sky.*
- "Intelligence" *is that which is for jews and not blacks.*

- **Lexical Definition Should Indicate the Context to Which the Definiens Pertains.**
- *Caveman* (prehistory) means a man who lives in the cave.
- *Caveman* (human behavior) means a crude person or a person without manner.



Fallacies in General

CHAPTER FIVE

Fallacies

- A fallacy in general can be defined *as an error in reasoning*. Moreover, fallacies can broadly be divided into **two types**, *formal and informal*. And, *informal fallacies*, which are the main focuses of this chapter, can be categorized into *five groups* considering some basic communality vested up on them. For that matter, fallacies of relevance, fallacies of weak induction, fallacies of presumption, fallacies of ambiguity, and fallacies of grammatical analogy are the main groups, which will seriously be considered in this chapter.

Cont'd

- Ordinarily speaking, fallacy can be understood as **false or mistaken belief**; however, logic perceives it **as a defect in the reasoning processes of arguments**. Thus, as logic dictates us, fallacy can broadly be understood as **a defect or mistake committed in arguments or processes of reasoning**. To further understand the notion of fallacy, it forces us to go back to the etymological origin of the word. For that matter, the term fallacy comes from the Latin “fallo”, “fallcia”, which signifies deception, trick or cheating.

- Fallacies can broadly be classified into two types: *formal* and *informal*. Though the focus of this chapter is on informal fallacies, it sounds logical to raise some preparatory notions about both of these classifications. For that matter formal fallacies are committed when the *form or logical structure of arguments are violated* whereas *informal fallacies are committed when the content of an argument is problematic*. Formal fallacies, which are committed because of *structural defects of arguments*, involve an explicit use of an *invalid deductive argument forms*. And, a deductive argument is invalid and fallacious formally usually because the premises fail to support the conclusion with **strict necessity** or when the premises of an argument are true and followed by a false conclusion.

Formal Fallacy

All tigers are animals

All mammals are animals

Therefore, all tigers are mammals.

All A are B

All C are B

Therefore, all A are C

- In this particular argument we do not have the ground to infer the above stated conclusion. The premises doesn't imply the conclusion of the above argument. We do not have the basis to argue that all tigers are mammals based on the given premises.

Example

- All factories are plants
- All plants are things that contain chlorophyll

Therefore, all factories are things that contain chlorophyll.

All A are B

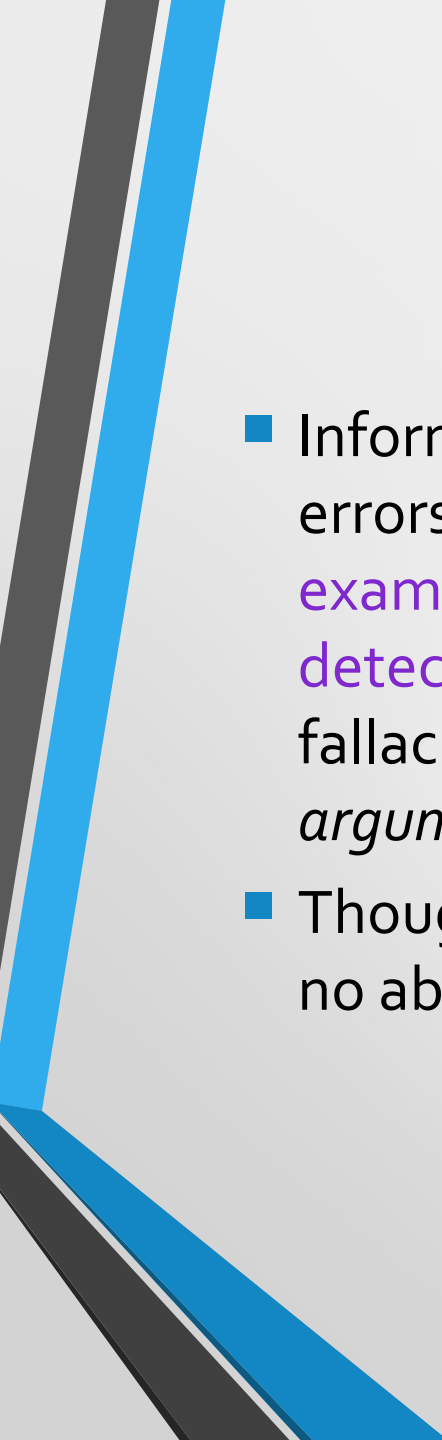
All A are B

All B are C

All C are D

Therefore, all A are C

Thus, all A are D

- 
- Informal fallacies, which are the focus of this chapter, are errors in reasoning which are detected through examining the **content** of an argument, not through detecting the form of an argument. And, informal fallacies may appear in *both **deductive and inductive** arguments*.
 - Though there are very many informal fallacies, there is no absolute consensus on how can they be classified.

❏ Fallacies of Relevance

- When an argument is based on premises, which are not relevant to its conclusion, and that therefore, cannot possibly establish its truth, fallacies of relevance will be committed. In other words, those fallacies, which are included under relevance, involve *premises, which are logically irrelevant to the conclusion, but for psychological reasons, they may seem relevant.* In any case, all fallacies of relevance commonly share the following basic features:

Appeal to Force

- This fallacy occurs or committed when an arguer imposes her/his conclusion or position through employing threats of force or any psychological intimidation in its premises so as to make the audiences or readers accept her/his conclusion. This is achieved through indicating that some danger will be happen on those who do not accept the position.

Examples:

- **Child to its Playmates:** *Arsenal is the best football club in the world, if you don't accept this, I am going to call my brother and he will throw you out!*
- **Wife to husband:** *I deserve a weekend in Langanò and if you don't agree to take me there; I am going to pack up the kids and leave.*

Secretary to Boss: I deserve a raise in salary for the coming year. After all, you know how friendly I am with your wife, and I'm sure you wouldn't want her to find out what's been going on between you and that sexpot client of yours.

Appeal to Pity (Argumentum ad Misericordiam)

- The word '*Misericordiam*' is originally from Latin which literally means 'a pitying heart' which in turn implies the request of someone to others so as to get mercy, sympathy or any a kind of excuse. Thus an appeal to pity fallacy basically occurs when an arguer tries to pose a conclusion by evoking pity from the listeners or readers. In other words, the fallacy is committed when the emotional appeal which raises the pity of the listeners or readers replace logical evidences or justifications.

Examples:

- **Attorney to a Judge:** *Members of the jury, I realize there is a good deal of evidence that these two brothers killed their parents. But they are now orphans. They have no one to take care of them. They must now face the cruel world afraid and alone. Surely they are not guilty of these heinous crimes.*
- **A Student to her Professor:** *Professor, it would be wrong for you to flunk me for cheating. I am a single mother, and to provide for my two kids. I have to work three jobs. At the end of the day, I am absolutely exhausted, and after I drag my weary body home. I have neither the time nor the energy to study.*

Appeal to People

- Naturally, every one of us wants to be accepted, loved, and esteemed by others. However, the problem lies on how (is that through appealing to emotion or through forwarding rational justifications or evidences) can we secure this desire.
- In other words, these public figures usually commit appeal to people fallacy since they forward premises with contents of emotive and expressive languages and devices so as to raise the Mob mentality of the crowds and make the crowds accept their side or conclusion.

Cont'd

- Thus to understand appeal to people fallacy, there are two approaches: ***direct*** and ***indirect approaches***. The fallacy will be committed directly when the arguer, addressing a large group of people through writing or speech, excites the emotions and enthusiasm of the crowds so as to win acceptance or to raise the mob mentality from them. As it has been stated before, political candidates for election, military leaders and other public figures usually employ propaganda so as to raise their subject and make them accept their conclusion

Example

- A leader of demonstration to his/her followers: *"Fellow citizens! Today we are threatened with the loss of our sacred to bear arms. A pack or gutless, liberal politicians wants to outlaw the sale of handguns. But they won't get away with it. The right to bear arms is guaranteed in the constitution! Our blessed forefathers created this right and bestowed it on all Americans. Protect the constitution! Hear our cry! Down with the gutless Pinkos!"*

Cont'd

- On the other hand, appeal to people fallacy is committed indirectly in the sense that the arguer *directs his or her appeal not to the crowd as a whole directly, rather to some aspects of their relationship to the crowd.*
- However there are **three** types of **indirect approach** to ad populum fallacy; ***appeal to bandwagon, appeal to vanity and appeal to snobbery.***
- **Appeal to Bandwagon** emphasizes that the majority's choice is the correct one and urges the audiences to join them. In other words, if some argue as you will be left behind or left out of the group/majority if you do not use the product.

Example

- Sure, trident is a very fantastic gum with lovely flavor. That is why the ***majority*** of the people in Addis Ababa chew it than any other gums.
- Chewing chat cannot be all wrong because 70% of students of Addis Ababa university have found nothing wrong with it.

Cont'd

- **Appeal to Vanity** is committed when an arguer associates products with celebrities and popular figures such as artists, athletes, footballers, etc. and informs the audiences that if they buy the item they will also be admired too.
- **Example:**

Appeal to Snobbery

- Committed when an arguer propounds the position that “if you want to be a member of the selected few you should use this product” i.e.-products are usually associated with persons with high social positions (Business man, Kings, queens, and princes).
- **Examples:** *A Rolls-Royce is not for everyone. If you qualify as one of the select few, this distinguished classic may be seen and driven at British Motor Cars, Ltd. (By appointment only, please.)*

Argument against the Person

- This fallacy occurs when, instead of responding to the argument forwarded by Mister 'X', Mister 'Y' tries to attack against Mister 'X' himself. Here, mister 'Y' clearly commits argument against the person fallacy.

Fallacy of ad hominem Abusive

- This is the fallacy committed when an arguer engages in direct personal attacks or abuses against his opponent and makes them as grounds to reject his claim. Thus, rejecting our opponent by directing our attack towards his personality rather than the contents of his argument will result in the fallacy concerned.

Examples

- *Ato Gebeyhu has argued for increased funding for the disabled. But nobody should listen to his argument. Ato Gebeyhu is a person who cheats on his wife, beats his wife and kids, and never pays his bills on time.*
- Media host Howard Stern claims that leaders of the religious right are nothing but a pack of racketeers bent on destroying every vestige of free speech, and he gives numerous reasons to support this claim. But Stern is just a vulgar, smut-mouthed freak who will say anything for shock value. Nobody should listen to this nonsense.

Fallacy of ad hominem Circumstantial

- This is the fallacy committed by an arguer who tries to discredit his opponent's arguments by alluding to certain circumstances that affect them (his opponents).
- **Example:** *Mr. Mohammed has just argued that we replace the public school system with private education. But of course he argues that way. He has no kids, and he does not want to pay any more taxes for public education.*

Tu quoque (you too) Fallacy

- **Example:** your argument that I should stop stealing candy from the corner store is no good. You told me yourself just a week ago that you, too, stole candy when you were a kid.

Accident

- This fallacy is committed when the general rule, principle or truth is wrongly applied to particular instance or situation.
- Examples: *Children should obey their parents. Therefore, little Abush should follow his alcoholic fathers orders to drop out of school and get a job.*

Cont'd

- People are obligated to keep their promises. When Jessica married Tyler, she promised to stay with him for life. Therefore, she should stay with him now, even though he has become an abusive spouse addicted to gambling and drugs.

Strawman

- This fallacy occurs when someone distorts his/her opponent's argument for the purpose of more easily attacking or demolishing.
- Examples: Mr. Goldberg has argued against prayer in the public schools. Obviously Mr. Goldberg advocates atheism. But atheism is what they used to have in Russia. Atheism leads to the suppression of all religions and the replacement of God by an omnipotent state. Is that what we want for this country? I hardly think so. Clearly Mr. Goldberg's argument is nonsense.

Cont'd

- *Dr. Kebede has just argued against affirmative action for women. It seems what he is saying is that women should stay out of the work place altogether. Just keep them barefoot and pregnant. That is what Dr. Kebede wants. Well! I think we are all smart enough to reject his argument.*

Missing the Point

- Missing the point illustrates a special form of irrelevance, which occurs when the premise of an argument supports the conclusion, which has nothing to do with correct conclusion.
- Examples: Crimes of theft and robbery have been increasing at an alarming rate lately. The conclusion is obvious: We must reinstate the death penalty immediately.

- 
- *Haile Gebresilassie has won many cross country championships. He is still dedicated, hard worker, disciplined, courageous and determined to win marathon. Therefore, Ethiopians should save their lives from HIV-AIDS.*

Red herring

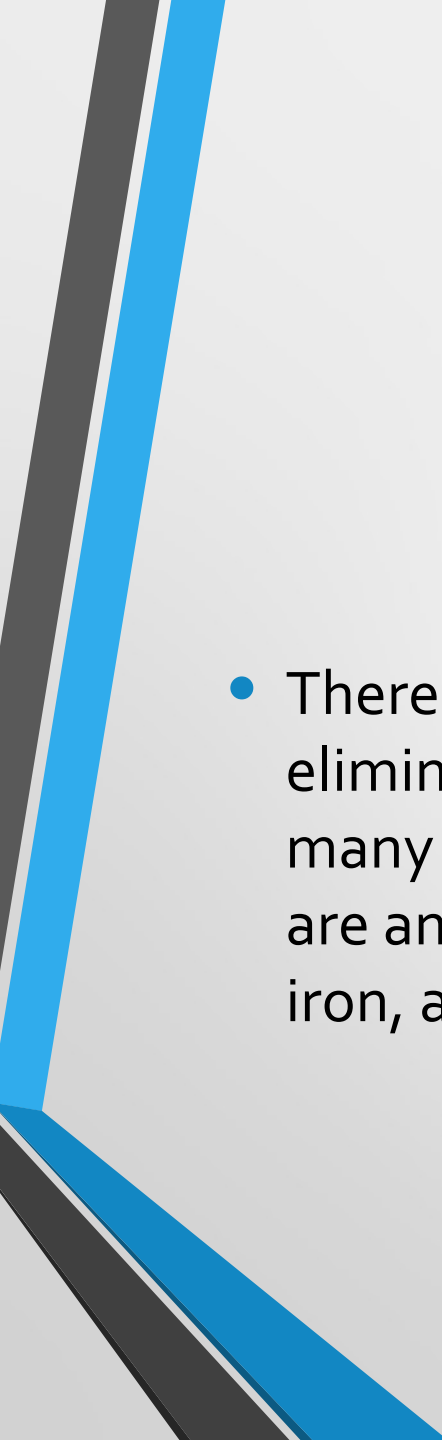
- Red herring fallacy will be committed when an arguer diverts the attention of the listeners or readers by changing the original subject to some totally different issue without the listeners or readers notifying it.
- An idea under discussion will be changed in to a totally different issue and then the conclusion will be drawn based on this changed subject.

Examples

- Professor Conway complains of inadequate parking on our campus. But did you know that last year Conway carried on a torrid love affair with a member of the English Department? The two used to meet every day for secret sex in the copier room. Apparently they didn't realize how much you can see through that fogged glass window. Even the students got an eyeful. Enough said about Conway.

Fallacies of Weak Induction

- Those fallacies included in weak induction occur not because the premises are logically irrelevant to the conclusion; rather it is because the **connection between the premises and conclusion is not strong enough**. Those fallacies under this category provide shared evidences to the conclusion. The evidences; however, are not good to make any reasonable person believe the conclusion.

- 
- There is a good deal of talk these days about the need to eliminate pesticides from our fruit and vegetables. But many of these foods are essential to our health. Carrots are an excellent source of vitamin A, broccoli is rich in iron, and oranges and grapefruits are high in vitamin C.

Appeal to Unqualified Authority

- Thus, the fallacy of unqualified authority is committed when we attempt to support our claim by:
- Citing the statement of another person who is not an authority in the field of specialization.
- Referring the judgment of an authority that is likely to be biased.
- Referring a person who has the habit of telling lies or disseminating wrong information.

Examples:

- *Omer, who is a well-known astronomer, says that AIDS epidemic is caused by a perverse alignment of the planets, and that there is nothing anyone can do about it. Therefore, we can only conclude that all of these efforts to find a cure for AIDS are a waste of time.*
- Old Mrs. Ferguson (who is practically blind) has testified that she saw the defendant stab the victim with a bayonet while she was standing in the twilight shadows 100 yards from the incident. Therefore, members of the jury, you must find the defendant guilty.

Appeal to Ignorance

- This fallacy is committed when the lack to evidence or proof for something is used to support the conclusion. In other words, when the premises of an argument state that nothing has been proved in one way or other about something and the conclusion then makes a definite assertion about that thing.

Examples:

- *Nobody has ever proved the existence of ghosts. Therefore, we have no alternative but to conclude that ghosts are mere figments of the imagination.*

Exceptional cases:

- Teams of scientists attempted over several decades to detect the existence of the luminiferous aether, and all failed to do so. Therefore, the luminiferous aether does not exist.
- Appeal to ignorance fallacy is correct in legal procedure; however, when a judge pronounces a verdict of no guilty, he/she is not claiming that the defendant did not commit the act as charged. It is only claiming that the evidence is not weightily enough to prove such a charge.

Hasty generalization

- The fallacy of hasty generalization, which is the opposite of accident, is committed when an arguer tries to generalize about a thing or an event based on insufficient evidence; very limited information and unrepresentative samples about it.

Examples:

- *A reporter in the local newspaper exaggerated her story just to make it more exciting, and a reporter on the evening news got her facts mixed up. The conclusion is apparent that you just cannot trust the news media today.*
- *Today's money managers are a pack of thieves, every last one of them. Look at Bernie Madoff and Robert Allen Stanford. They ripped off billions of dollars from thousands of trusting clients. And Raj Rajaratnam profited to the tune of millions of dollars through illegal insider trading.*

Exceptional cases:

- However, the mere fact that a sample may be small does not necessarily guarantee that the fallacy of converse accident occurs. Other factors sometimes make the argument to be strong though its samples are small.

Example:

- *Ten Milligrams of Substance Z was fed to four mice, and within two minutes all four went into shock and died. Probably substance Z, in this amount, is fatal to the average mice.*

False cause

- False cause fallacy occurs whenever the link between premises and conclusion depends on some imagined causal connection that properly does not exist, an attempt to suppose that 'X' causes 'Y' where as 'X' probably does not cause 'Y' at all.

And, the fallacy can further be divided in to three types:

A. Post hoc ergo propter hoc fallacy

B. Non Causa pro causa fallacy

C. Oversimplified cause fallacy

- A. The fallacy is shortly named as post hoc fallacy and it is committed when we arrive at a certain conclusion by claiming that one thing is the cause of another thing because it precedes in time
- Examples:

During the past two months, every time that the supporters have worn the blue shirt, the Ethiopian national football team has been defeated. Therefore, to prevent defeats in the future, the supporters should get rid of those blue shirts.

- *Every time I take a shower, the telephone rings. Therefore, since I'm dying to talk to somebody right now, I should jump in the shower.*

- **Non Causa pro causa fallacy**

The Latin phrase ***non causa pro causa*** can be translated as '***not the cause for the cause***'. The fallacy is committed when someone argues that something is the cause of an effect when it is not in reality and confusion occurs between cause and effect.

Examples:

- *There are more churches in Ethiopia today than ever before and more HIV victims ever before; so, to eliminate the epidemic we must abolish the church.*
- *The quality of education is declining at an alarming rate. We must therefore question the capacity of our teachers.*

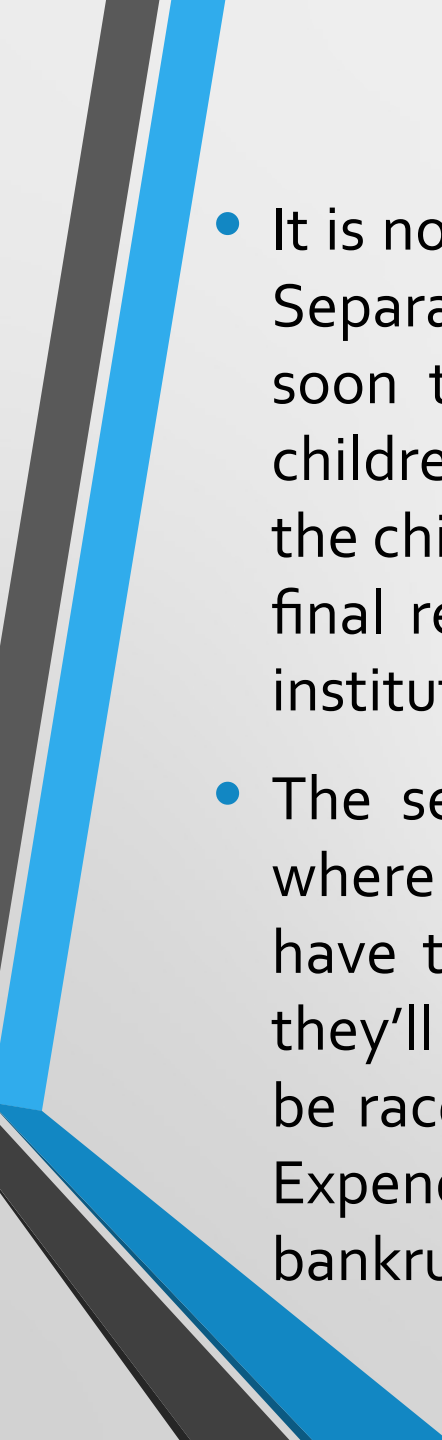
- Oversimplified fallacy

Oversimplified cause fallacy is committed when relevant causal antecedents of an event are oversimplified by introducing factors insufficient to the account of the effect.

- Examples: *Why most students fail in logic is because teachers do not come to class regularly.*
- The quality of education in our grade schools and high schools has been declining for years. Clearly, our teachers just aren't doing their job these days.

Slippery Slope

- If an arguer assumes that series of events happen or follow one from the other as a result of the first cause in a series, it will result in slippery slope fallacy. This is of the fact that because it is logically mistaken for someone to consider a particular action or event (usually the first one) in series of events causes for series of consequences.

- 
- It is not a good idea to put your child in a day care center. Separation from parents causes isolation and alienation soon the child becomes incapable of relating to other children, and this inability to relate causes depression. As the child gets older, the depression leads to psychosis. The final result is either suicide or a life wasted in a mental institution.
 - The secretaries have asked us to provide lounge areas where they can spend their coffee breaks. This request will have to be refused. If we give them lounge areas, next they'll be asking for spas and swimming pools. Then it will be racquet ball courts, tennis courts, and fitness centers. Expenditures for these facilities will drive us into bankruptcy.

- **Fallacy of Weak analogy**

Fallacy of weak analogy is committed basically when the analogy or similarity between two things or situations is not strong enough to support the conclusion to be drawn. In other words, weak analogy fallacy will be committed when the significant differences between two or more things compared are ignored or when two contrasted things are considered alike only in unimportant ways.

Examples:

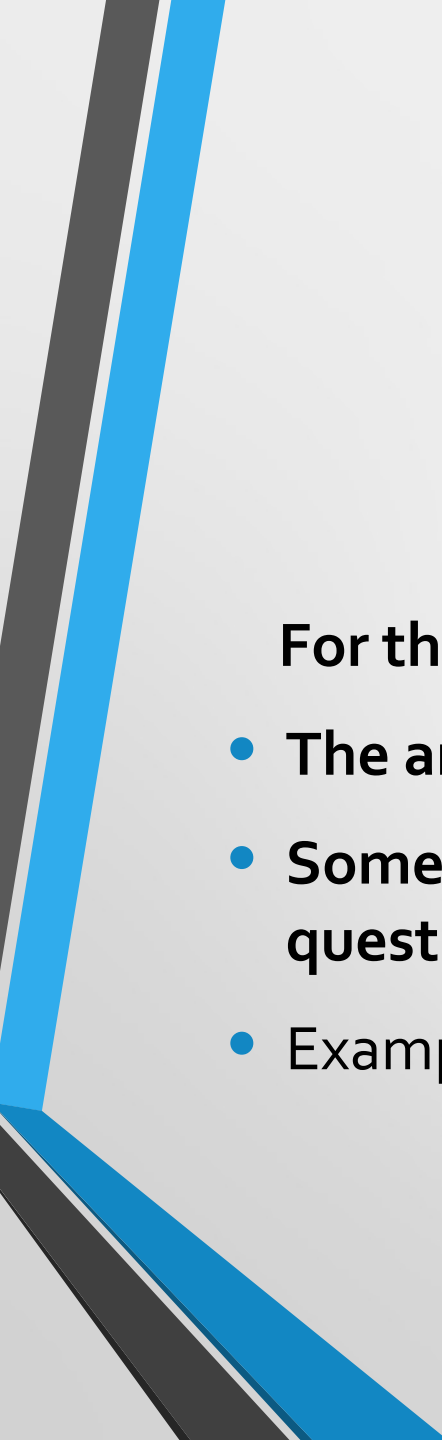
- *No one would buy a pair of shoes without trying them on. Why should anyone be expected to get married without premarital sex?*
- The flow of electricity through a wire is similar to the flow of water through a pipe. When water runs downhill through a pipe, the pressure at the bottom of the hill is greater than it is at the top. Thus, when electricity flows downhill through a wire, the voltage should be greater at the bottom of the hill than at the top.

Fallacies of Presumption

- To presume means to take something for granted or to assume a given idea as true or correct which in fact needs further proof, explanation or evidence. And, the fallacy of presumption will be committed when the assumption given in the premise is not supported by proof but the arguer maintains that it does not need proof and s/he invites his/her audiences accept it as it is.

Begging the question

- Begging the question fallacy basically occurs when someone uses some form of phraseology, which tends to conceal the questionably true character of a key premise, and, if the audience is deceived into this argument as sound, when in facts it many not be.



For this fallacy to be committed:

- **The argument must be valid,**
- **Some form of phraseology must be used to conceal the questionably true character of a key premise.**
- **Examples:**

Examples:

- Capital punishment is justified for the crimes of murder and kidnapping because it is quite legitimate and appropriate that someone be put to death for having committed such hateful and inhuman acts.
- Murder is morally wrong. This being the case, it follows that abortion is morally wrong.

Complex Question

- One commits the fallacy of complex question when s/he asks two or more questions in a way that makes it appear that only one question has been asked. When we forward question to someone we make presuppositions of answers within it.

Example:

- *Have you stopped cheating on exams? Yes.*
 - Are you cheating on exams? Yes
 - Were you cheating on exams? Yes

False dichotomy

- Thus, one commits the fallacy of false dilemma when he poses a restrictive set of undesirable alternatives when other legitimate alternatives may be possible. And, the fallacious nature of false dichotomy lies in the attempt by the arguer to delude the reader or listener into thinking that the disjunctive premise presents jointly exhaustive alternatives, and is therefore true by necessity.

Examples:

- *Either we elect EPRDF, or the country's fate will be worsened. The choice should be obvious.*
- Either you buy only Ethiopian-made products or you don't deserve to be called a loyal Ethiopian. Yesterday you bought a new Toyota. It's therefore clear that you don't deserve to be called a loyal Ethiopian.

Cont'd

- Either you marry me or I will commit suicide. I know you don't want me to commit suicide. Therefore, you should marry me

Suppressed Evidence

- This is an inductive argument that overlooks an important piece of evidence (premise) that if it were considered, the conclusion to be drawn would be very different from the one drawn. In other words, this fallacy is committed when the argument ignores some important evidence/s that outweigh/s the presented evidence and entails a different conclusion.

Example:

- *Addis Ababa University deserves to be one of the best Universities in Africa as it has impressive buildings, beautiful gates, and an attractive fountain.*
- Most dogs are friendly and pose no threat to people who pet them. Therefore, it would be safe to pet the little dog that is approaching us now.

Fallacies of Ambiguity

- Two fallacies arise from the occurrence of some form of ambiguity in either the premise or the conclusion (or both). The fallacies of ambiguity include Equivocation and Amphiboly.

Equivocation

- The fallacy of equivocation occurs when a word or phrase in a single argument are used in two different senses or connotations which in turn leads to a conclusion which is not supported by its premises. In other words, this fallacy leads toward unintended conclusion by making a word or words to have two different meanings in a single argument.

Example:

- *Some triangles are **obtuse**. Whatever is **obtuse** is ignorant. Therefore same triangles are ignorant.*
- Any **law** can be repealed by the legislative authority. But the **law** of gravity is a law. Therefore, the law of gravity can be repealed by the legislative authority.

Amphiboly

- The fallacy of amphiboly occurs when someone misinterprets a statement which is ambiguous because of some structural defects and draws a conclusion based on such misinterpretation.
- Moreover, the structural defect is usually a mistake in grammar, punctuation, a pronoun, an ambiguous antecedent of a pronoun, careless arrangements of words and the like

Example:

- John told Henry that he had made a mistake. It follows that John has at least the courage to admit his own mistakes.
- Mr. James signed a contract that reads, "In exchange for painting my house, I promise to pay David \$5,000, and give him my new Cadillac only if he finishes the job by May 1." Therefore, since David did not finish until May 10, it follows that he gets neither the \$5,000 nor the Cadillac.

Fallacies of Grammatical Analogy

Composition

- The erroneous transference of an attribute from parts of entity onto the whole as a base for the conclusion of an argument will lead to the fallacy of composition to occur.
- Example: *Each atom in this table is invisible to the naked eye. Therefore, the table is invisible to the naked eye.*

Cont'd

- Sodium and chlorine, the atomic components of salt, are both deadly poisons. Therefore, salt is a deadly poison.
- *Each atom in this piece of chalk has mass. Therefore, the piece of chalk has mass. (No fallacy)*

Collective and general statement

Fleas are small.

Fleas are numerous.

- The first statement is a **general statement**. The attribute of **being small** is predicated distributively; that is, it is assigned (or distributed) **to each and every flea** in the class. Each and every flea in the class is said to be small.

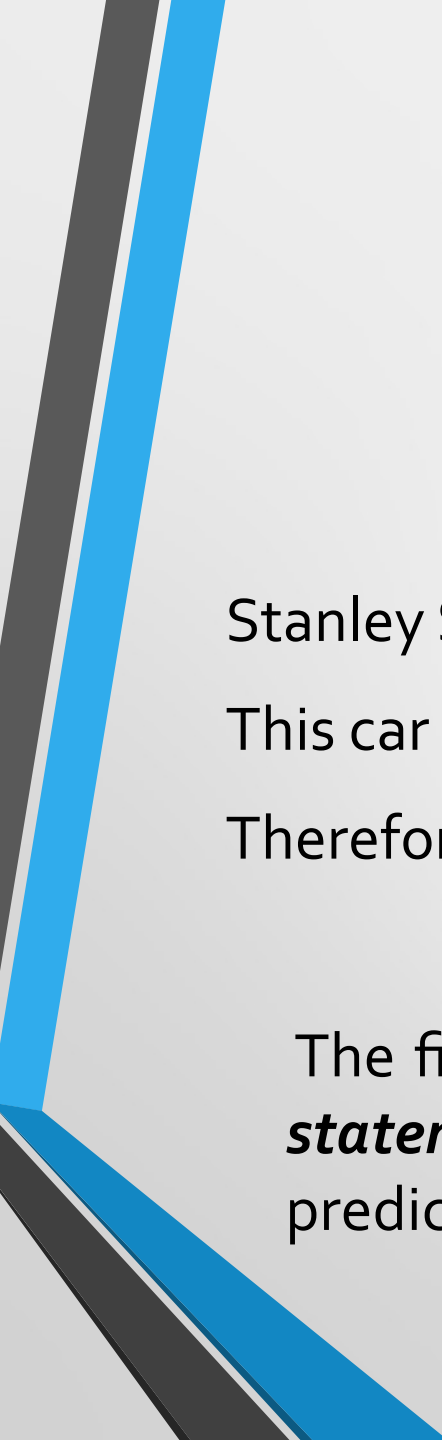
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- The second statement, on the other hand, is a statement about a class as a whole, or what we will call a “class statement.” The attribute of ***being numerous*** is predicated ***collectively***; in other words, ***it is assigned not to the individual fleas but to the class of fleas.***

Division

- The fallacy of division is the exact opposite of composition. As composition goes from parts to whole, division goes from whole to parts. The fallacy occurs when the conclusion of an argument depends on the erroneous transference of an attribute from a whole (a class) onto its parts (members).

Example:

- Salt is a nonpoisonous compound. Therefore, its component elements, sodium and chlorine, are nonpoisonous.
- The Royal Society is over 300 years old. Professor Thompson is a member of the Royal Society. Therefore, Professor Thompson is over 300 years old.



Stanley Steamers have almost disappeared.

This car is a Stanley Steamer.

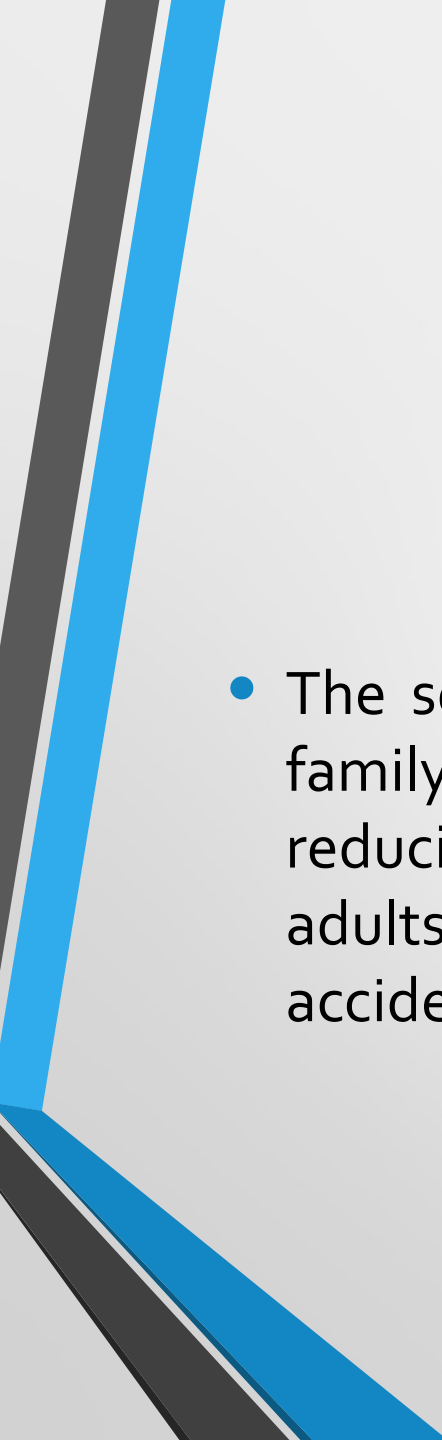
Therefore, this car has almost disappeared.

The first premise ***is not a general statement but a class statement***. The attribute of having almost disappeared is predicated collectively.

- 
- *The average American family has 2.5 children.*
 - *The Jones family is an average American family.*

Therefore, the Jones family has 2.5 children.

- The statement ***"The average American family has 2.5 children" is not a general statement, but rather a class statement.***

- 
- The sense of the statement is not that each and every family has 2.5 children, but that the class of families is reducible to about 55 percent children and 45 percent adults. Thus, once again, the fallacy is division, and not accident.